**3.**

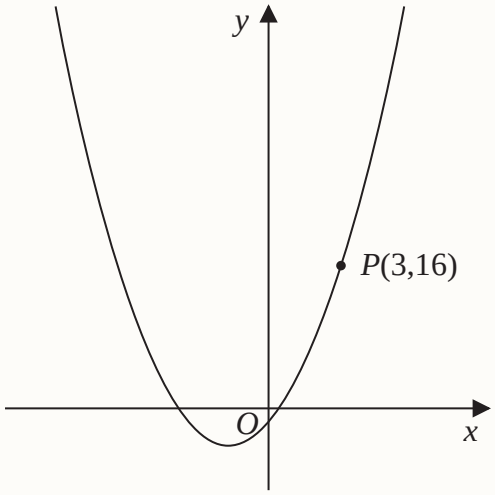



**Figure 1**

Figure 1 shows part of the curve with equation $y = x^2 + 3x - 2$

The point $P(3,16)$ lies on the curve.

(a) Find the gradient of the tangent to the curve at $P$.

**(2)**

The point $Q$ with $x$ coordinate $3 + h$ also lies on the curve.

(b) Find, in terms of $h$, the gradient of the line $PQ$. Write your answer in simplest form.

**(3)**

(c) Explain briefly the relationship between the answer to (b) and the answer to (a).

**(1)**